

120 Days of Quantitative Finance

Week 8: Volatility Models & Returns Forecasting

Day 54: Financial Applications – Value at Risk (VaR)

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1. Definition of Value at Risk

Value at Risk (VaR) at confidence level α is defined as:

$$P(Loss > VaR_\alpha) = 1 - \alpha$$

where loss is:

$$L_t = -r_t$$

2. Parametric VaR

Assuming returns are normally distributed:

$$r_t \sim N(\mu, \sigma^2)$$

Then:

$$VaR_\alpha = -(\mu + z_\alpha \sigma)$$

where z_α is the standard normal quantile.

3. Historical Simulation VaR

$$VaR_\alpha = -\text{Quantile}_{1-\alpha}(r_t)$$

This method makes no distributional assumptions.

4. Monte Carlo VaR

1. Model return distribution.
2. Simulate returns.
3. Compute loss distribution.
4. Extract quantile.

5. Expected Shortfall

Expected Shortfall (ES) is:

$$ES_{\alpha} = E[L \mid L > VaR_{\alpha}]$$

It measures the average loss in the worst α percent of cases.

6. Conditional VaR with GARCH

If:

$$r_t = \sigma_t z_t$$

then conditional VaR becomes:

$$VaR_{t+1} = -z_{\alpha} \sigma_{t+1}$$

where σ_{t+1} is obtained from volatility forecasting models.

Conclusion

VaR is a central risk management tool used by financial institutions. However, it has limitations, and Expected Shortfall is often preferred for regulatory and tail-risk measurement purposes.